

AMAN ADHAV

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Machine Learning Engineer | NLP & Recommender Systems | 1st of 50, Amazon Kiro Challenge | Available Jan 2027

EDUCATION

Arizona State University

Bachelor of Science in Computer Science, GPA 3.83/4.0

Tempe, AZ

Expected Graduation: Dec 2026

Amazon Kiro Challenge: Grand Prize & Best in Build (\$6,000, 1st of 50 ASU teams)

Certifications: AWS Certified AI Practitioner, AWS Certified Cloud Practitioner | AWS Developer Associate (in progress)

EXPERIENCE

ASU Enterprise Technology

Software Engineer, Integration & APIs Team

Scottsdale, AZ

Mar 2026 – Present

- Built Python/AWS Lambda data pipelines feeding 80,000+ student records from Salesforce, Oracle ODS, and PostgreSQL to downstream ML and analytics consumers.
- Built an Athena querying skill opening an 80,000+ record event archive to analysis, validating field stability across large samples before fields are trusted downstream.
- Reclaimed roughly 15 engineer-hours per month by shipping LLM agents that automate event-source onboarding across Jira, Terraform, and GitHub.

Arizona State University

Undergraduate Teaching Assistant, Linear Algebra & Calculus

Tempe, AZ

Aug 2024 – Mar 2026

Rachita Infotech

Data Science Intern (2023); Software Engineering Intern (2024)

Pune, India

Summers 2023, 2024

- Shipped a customer churn prediction model at ~84% held-out accuracy using scikit-learn and XGBoost, then packaged it as a Dockerized Flask endpoint (2023).
- Cut data preparation time ~20% by cleaning and preprocessing large datasets in pandas and SQL, and surfaced behavioral trends through EDA and Tableau dashboards (2023).
- Improved internal project-tracking workflows ~15% with a full-stack React/Node/Express app, raising core-module Jest coverage from ~60% to ~75% (2024).

PROJECTS

TakeMeter, Fine-Tuned NLP Classifier | DistilBERT, Hugging Face

github.com/amanadhav/ai201-project3-takemeter

- Beat a zero-shot Llama 3.3 70B baseline by 30.6 points (69.4% vs 38.9% accuracy, macro F1 0.68) by fine-tuning DistilBERT for 3-class discourse classification.
- Built the 239-example training set from 615 scraped posts via pre-labeling, manual review, and stratified splits; diagnosed the top failure mode by error analysis.

CineMind, Hybrid Recommendation Engine | Python, scikit-learn, SciPy, Flask, Next.js

cinemind-prod.vercel.app

- Served recommendations over 100k+ MovieLens ratings (610 users x 9,724 movies) by blending k=50 SVD matrix factorization with TF-IDF cosine-similarity KNN over metadata.
- Solved cold-start without retraining by folding new ratings into the latent space, served via an 8-endpoint Flask API with 71 pytest tests.

Provenance Guard, AI Text Attribution | Python, Llama 3.3

github.com/amanadhav/ai201-project4-provenance-guard

- Improved short-text accuracy by weighting an LLM classifier against stylometric features adaptively by word count, after stylometrics over-scored short inputs.
- Cut false accusations against human writers via an asymmetric 0.71 AI-verdict threshold, an SQLite decision audit log, and an appeals workflow.

TECHNICAL SKILLS

ML & NLP: PyTorch (via Hugging Face), Hugging Face Transformers/Datasets, scikit-learn, SciPy, pandas, NumPy, DistilBERT, fine-tuning, transfer learning, text classification, TF-IDF, SVD/matrix factorization, collaborative filtering, recommender systems

Evaluation: baselines, precision/recall, macro F1, confusion matrices, stratified splits, error analysis, data labeling

LLM & Applied AI: RAG, ChromaDB, embeddings, semantic search, grounded generation, prompt engineering, Claude API, OpenAI API, agent orchestration, MCP, Gradio

Languages & Databases: Python, SQL, JavaScript, Bash, PostgreSQL, SQLite

Libraries & Frameworks: Flask, FastAPI, Pydantic, pytest, ETL pipelines

Cloud Infrastructure & Deployment: AWS (Lambda, S3, Athena, RDS), Terraform, Docker, Git